

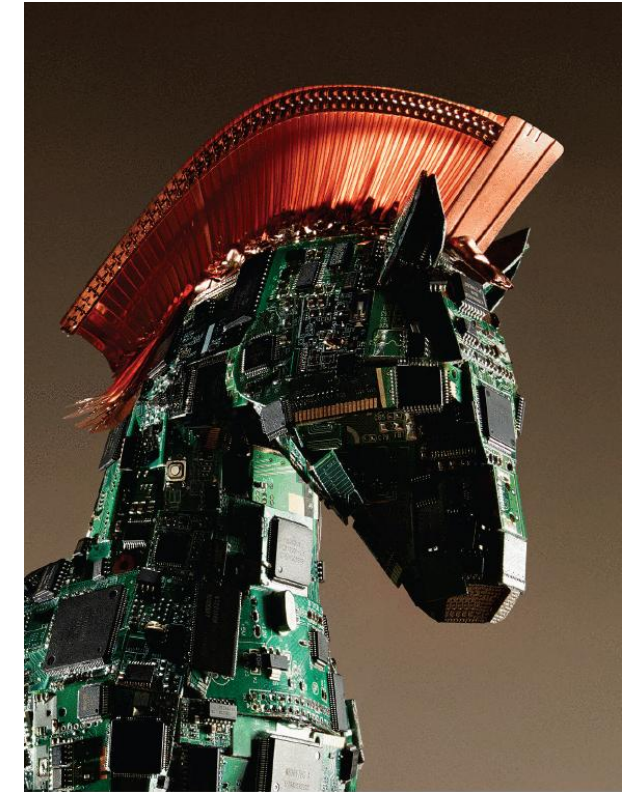
Community-Driven Benchmarking for Trojan Insertion versus Layout Defenses in Modern ICs

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Context

- Large-scale, 1st-of-its-kind community effort for red-versus-blue teaming of hardware Trojan (HT) insertion versus layout defenses
- Over several months, 4 independently competing blue teams tackled vulnerabilities of benchmarks layouts, whereas a red team utilized real HTs and explored novel insertion technique
- Real-world settings and constraints for a modern 7nm IC node and industry-grade tooling, all embedded into a fully-automated and extensible benchmarking framework
- Full release of codes and artifacts: <https://github.com/DfX-NYUAD/Trojan-Insertion-versus-Layout-Defenses>



IEEE Spectrum, 2015

Components

1) Technology Setup: ASAP7 PDK and Library

- Open source, originally developed by Arizona State and ARM
- Mimics real-world IC technology quite well
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2) Reference Flow for IC Design

- Follows prior experience and best practices from tapeouts; based on commercial tool setup
- Flow itself open-source (scripts) but tooling is closed access
 - All interested teams already had their own access, but still limiting factor for community at large
 - Next time: open-source flows like OpenROAD

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4) Further Artifacts

- Benchmarks: layouts synthesized from open-source crypto cores
- Trojans: RTL logic
- Attack and defense techniques: scripts
- Evaluation and scoring: scripts
- Results: layouts and reports

Challenges

Reproducibility: different tool versions provided different results

- Known and common issue in industry, especially when pushing design frontiers
- Fixed reference version enabled consistency for official evaluation
 - Still, frustrating for teams to close the (often erratic) gap b/w their internal and official results
- Not sure if open-source offerings are more stable; again, pushing design frontiers
 - But, should be easier to setup fixed version at teams' ends, whereas commercial tooling often fixed by department IT, other projects, etc.

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Complexity

- IC design for modern technology
- Different tools and data modes
- Novel techniques; iterative code revisions
- Full automation; requires extensive testing

Trojan Insertion versus Layout Defenses for Modern ICs: Red-versus-Blue Teaming in a Competitive Community Effort

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